

Things You May Like to Know

About Soldering



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How should an engineer go about adopting the right soldering method? This is probably a question that most electronics engineers face at least once in their lives. The first thing that needs to be done is selecting the soldering method best suited to an application by considering the pros and cons of each soldering method, as well as the heat resistance of the surface-mount device.

Soldering methods are broadly divided into the partial-heating method and the total-heating method. While heat is applied to the components' leads and printed wiring boards (PWBs) in a localised manner in the former, the latter sees heat being applied to the entire board. This means that partial heating involves less heat stress on the device and PWB, but is unsuitable for large-volume production. Therefore this method is mainly used to correct soldering or for devices with low heat resistance.

The total-heating method is excellent in terms of productivity and running cost, but it can place considerable heat stress on the semiconductor device and board. In both cases, engineers should learn to validate the process with respect to the job they intend to carry out.

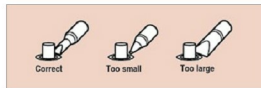


Fig. 1: Selecting the right soldering tip

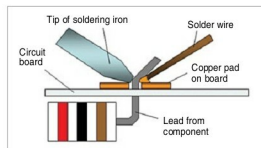


Fig. 2: Correct positioning of the solder tip

Partial-heating method: Hand soldering

Selecting the right soldering tip and iron.

This is something that needs to be practically validated according to the nature of the job and components used. The job dimension and tip dimension, which is about 0.5mm to 5.0mm wide, with different types of shapes like C type (round chamber shape) or D type (flat screwdriver type), need to be selected. Proximity of other components also needs to be considered. See Fig. 1 for clarity.

Thermal effects. Controlling the soldering iron tip temperature using the controls available on the soldering equipment is not the key element in soldering. The key element is in controlling the heat cycle of the work in terms of how fast it gets hot, how hot it gets and how long it stays hot for reliable solder connections.

Engineers should also learn about process validation of the soldering process with respect to job dimension, tip dimension, solder-wire dimension and temperature. Based on validation results and the operator's speed of performing soldering, temperature should be set between the minimum and maximum range of validated temperature for a particular job.

Maintaining the soldering tips. The optimum temperature at which proper soldering takes place should be set so that overheating is avoided. When the tip is cleaned on the sponge, the sponge should have only moisture; water should not drop from the sponge while squeezing it and you should only feel the moistness. Hence, it should be moistened to that level periodically.

Solder tips should not be rubbed or sharpened with a hard surface nor should these be tapped on the table. The handle set must always be kept with the tip in the stand when not in use, preferably with the tip covered in solder. Almost every day,